

Active Pointing Stabilization Techniques Applied to the Low Energy RHIC Electron Cooling Laser Transport at BNL

Linh Nguyen, Brookhaven National Laboratory, Upton, NY, USA

Co-authors: A. Curcio, W. Eisele, A.V. Fedotov, A. Fernando, W. Fischer, P. Inacker, J. Jamilkowski, D. Kayran, K. Kulmatycki, D. Lehn, T. Miller, M. Minty, A. Sukhanov, Brookhaven National Laboratory, Upton, NY, USA

NAPAC2019

NORTH AMERICAN PARTICLE ACCELERATOR CONFERENCE

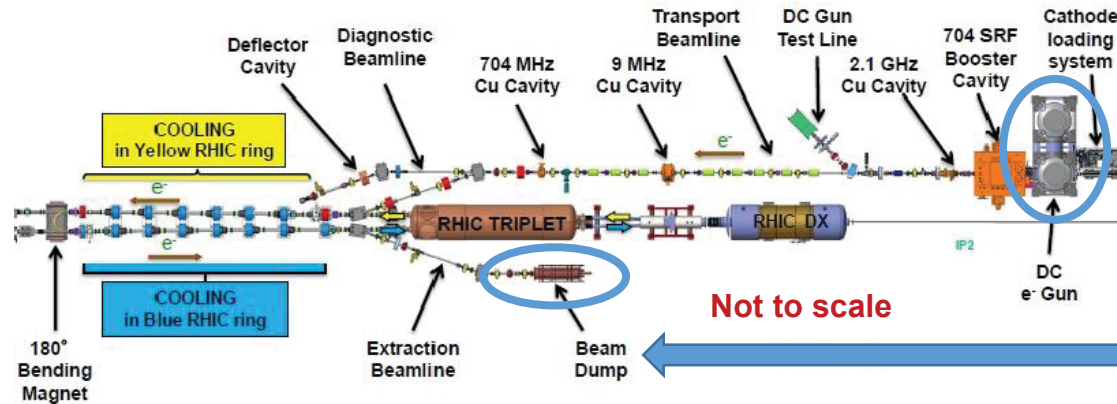
BROOKHAVEN
NATIONAL LABORATORY



BROOKHAVEN SCIENCE ASSOCIATES

The Low Energy RHIC electron Cooler (LEReC)

- First electron cooler using bunched electron beams, successfully commissioned in 2018 (TUZBA1)
- Demonstrated cooling of Au ions in the Relativistic Heavy Ion Collider (RHIC) at BNL this year (THZBA5)



Beam is generated by a high-power fiber laser illuminating a photocathode

Beam travels 100 m before reaching dump



- To remain accessible during operations, the laser is located outside the accelerator tunnel
- Output beam is propagated over a total distance of 34 m to the photocathode
→ Three laser tables are involved (trailer, relay, gun)
- A limit of 10 μm rms position variation was placed on the center-of-mass (CoM) of the laser beam spot on the photocathode

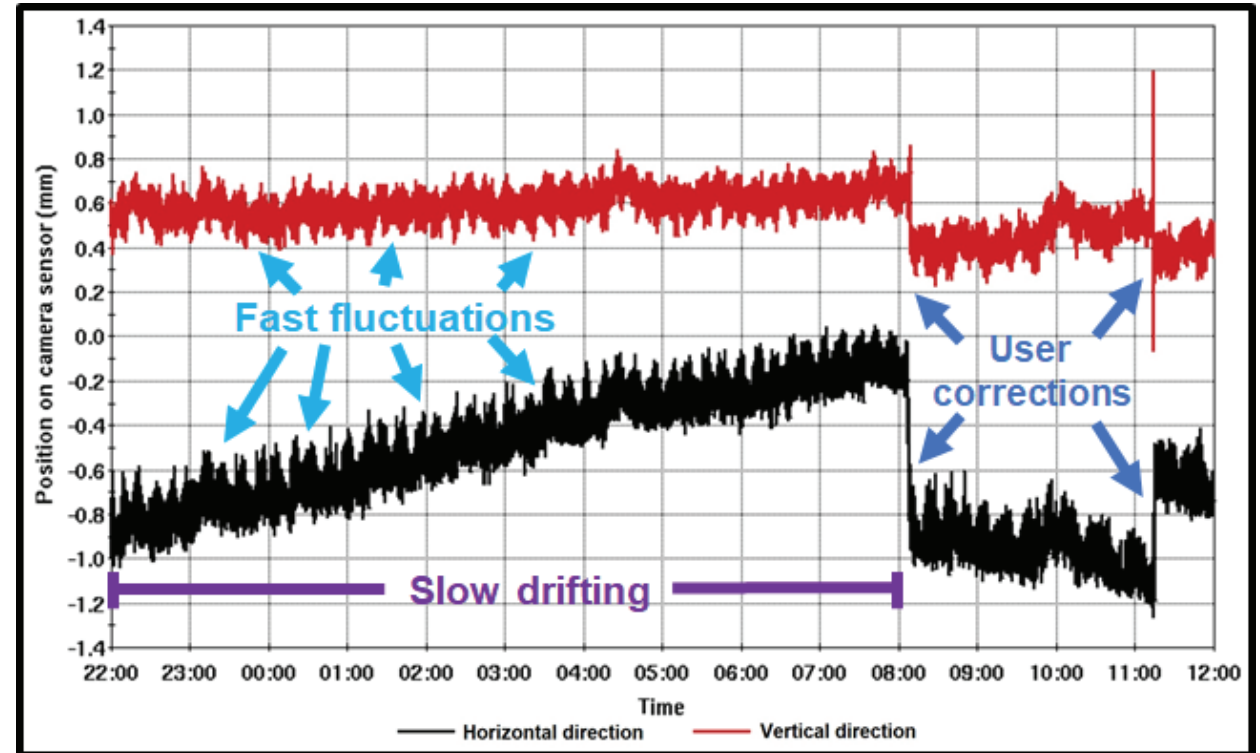
Fast Fluctuations vs. Slow Drifting

The 10 μm rms was not achieved during the 2018 Run.

- Passive stabilization not enough
- Need active stabilization

Two types of unwanted CoM movement:

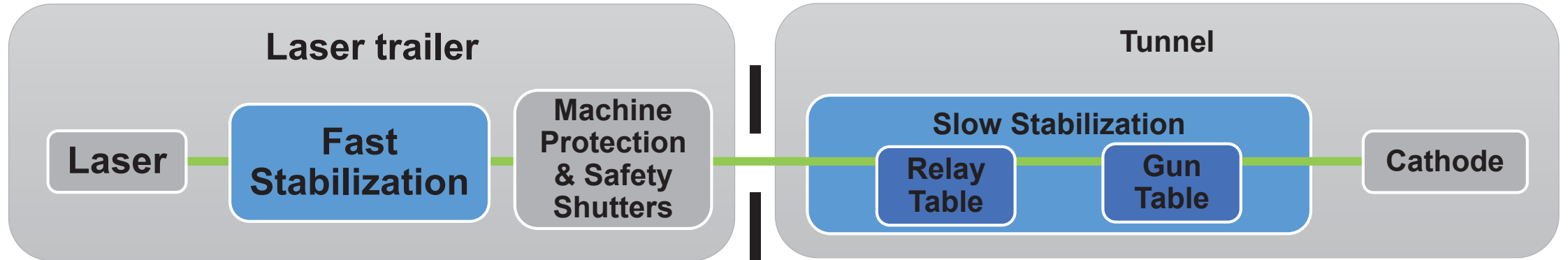
- **“Fast fluctuations”**
 - Shot-to-shot variations
 - Originate in the drive laser components
 - Compounded by air currents prior to the beam’s injection into the transport
- **“Slow drifting”**
 - Occurs over the course of hours
 - Arises from weather- and season-related relative movement of the three laser tables composing the transport



Types of CoM movement without active stabilization.
Data from operations during 2018 Run.

Laser pointing stabilization feedback was developed to mitigate position fluctuations and drifting

Approach

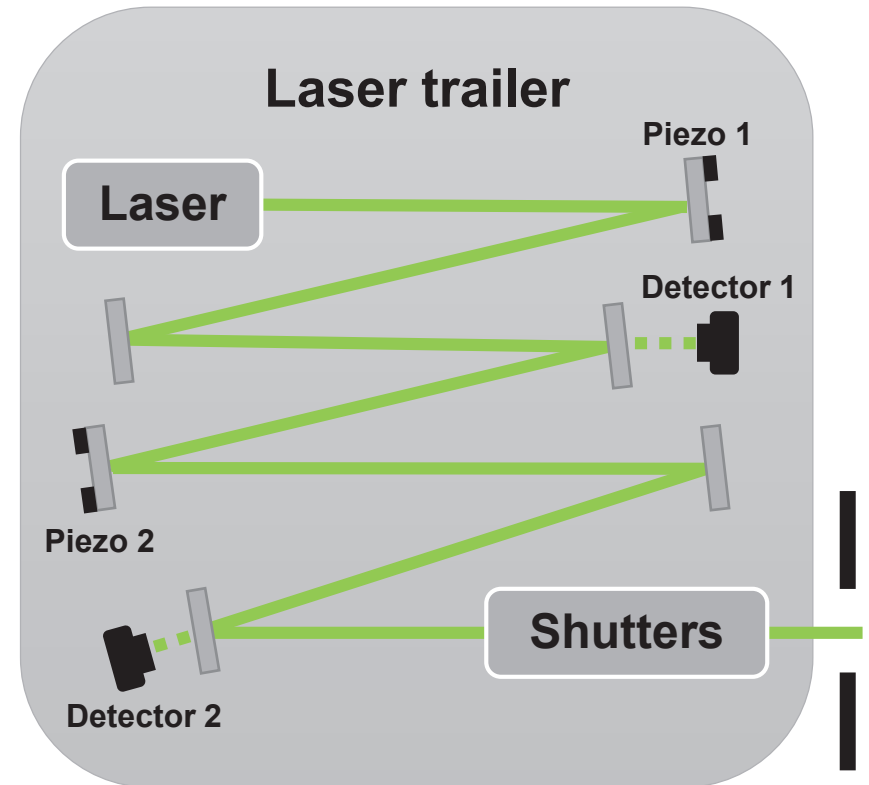


Conceptual layout of active stabilization techniques for LEReC

- Existing commercially available systems lack required flexibility (e.g., exception handling) for full transport stabilization
→ **Can be used to address the fast fluctuations present in the laser before the beam transport**
- To address the slow drifting of the laser beam through the transport, **in-house solution** was developed around the unique needs and infrastructure of LEReC and the RHIC complex

Fast Stabilization

- Uses stock active laser beam stabilization system (MRC)
- Stabilization section was inserted just before the transport
- Two serial detector-piezo mirror pairs
- Paired detector and actuated piezo mirror are separated by 1 m of free-space propagation and connected through a closed-loop controller
- Controller continuously adjusts the trajectory of the beam, removing fast fluctuations present in incoming beam
- Output beam has fixed position and angle
- Offers up to 100 kHz of bandwidth
- Uses leakage light with a limited dynamic range
 - Can only provide stabilization in CW mode
- Functionality in the manufacturer's controller enables remote activation and de-activation of the system
 - Pulsed mode can be programmatically avoided

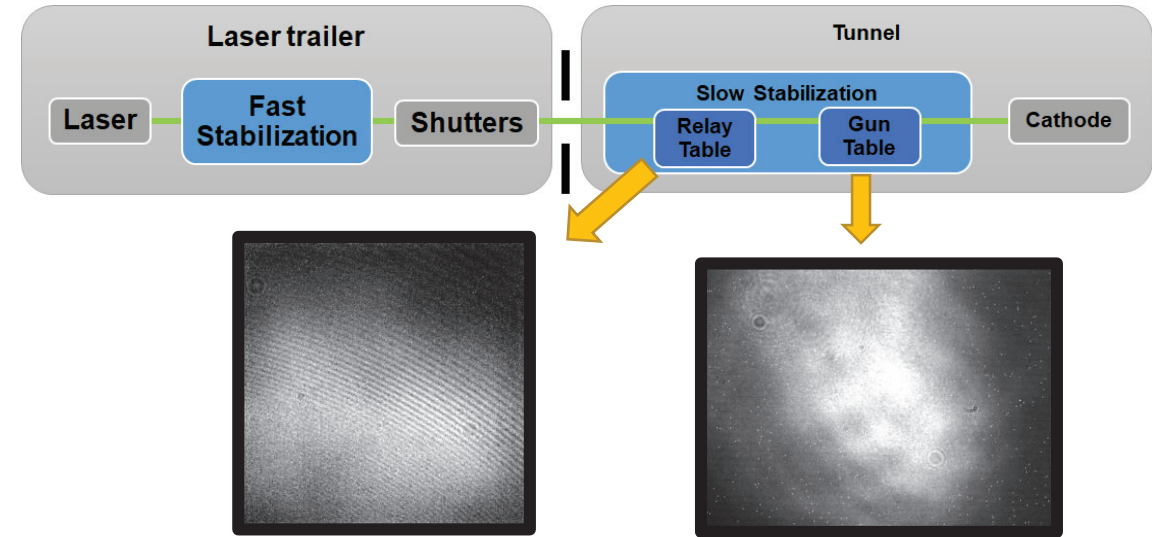


Conceptual layout of fast stabilization section.
Each detector is separated from its paired piezo mirror by about 1 m.

Slow Stabilization

- No longer necessary to sacrifice dynamic range and programming flexibility for speed in this regime
 - Slow, camera-based stabilization system using aligned reference frames was developed
 - By automatically adjusting camera settings and using flip filters, system achieves dynamic range to cover all operating modes without user input
 - System uses same steering mirrors as those used by operators to control the laser trajectory
- Two sets of cameras and flip filters were installed along the transport, fixing both position and angle
- Independent triggering and setting of camera parameters enable programmatic camera control (the so-called “dynamic camera loop”) for exception handling and ensuring image quality

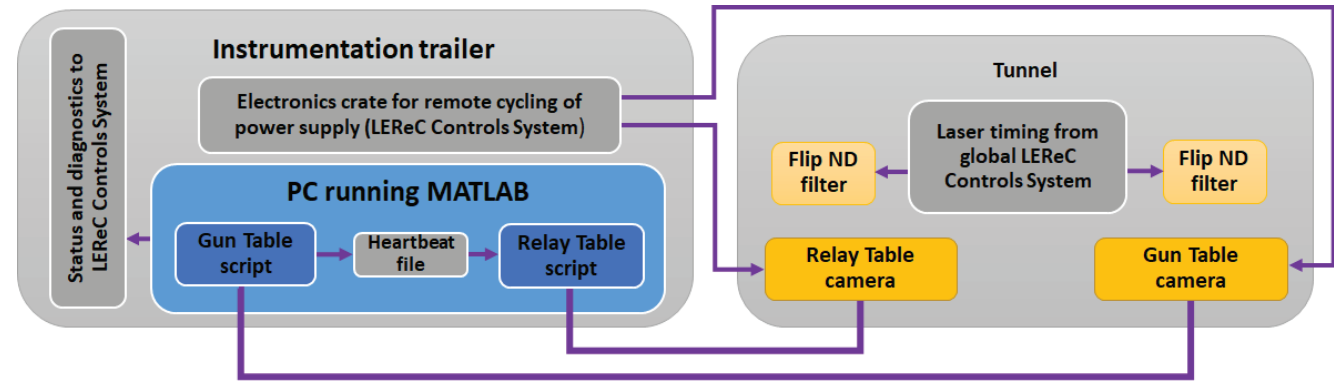
Conceptual layout of active stabilization techniques for LEReC



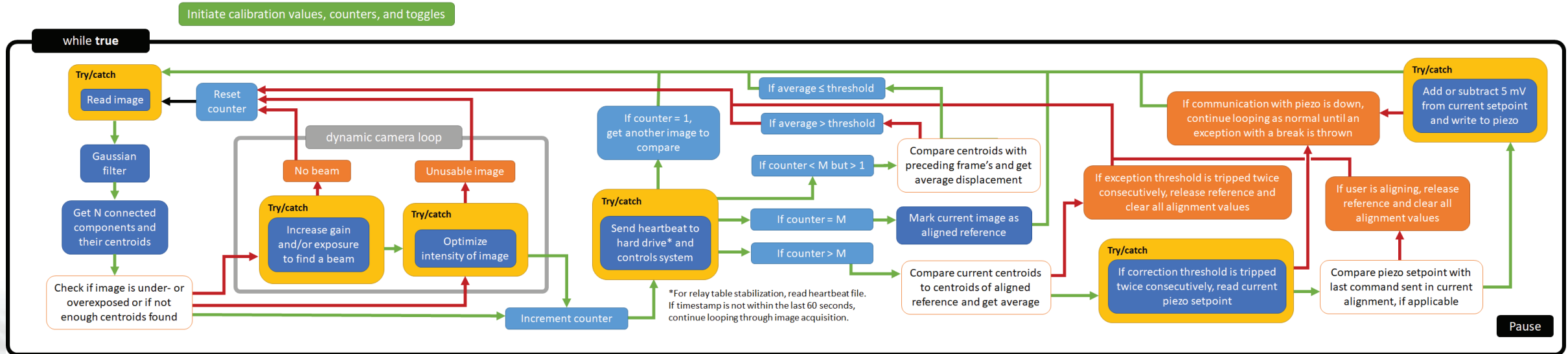
Example of laser beam profile on the relay table (left) and gun table (right) for the same beam through the transport in pulsed mode. The slow stabilization system was designed to handle both the relay table's large, diffuse beam and the gun table's highly structured beam with pulsed-to-CW (10^6) dynamic range.

Camera-Based System and Slow-Feedback Algorithm

- Two nearly identical MATLAB scripts run in parallel but at different bandwidths (gun table is at least 2x faster)
- Gun table stabilization must be active for relay table stabilization to become active, but systems are otherwise independent
- Algorithm determines a reference frame with which to compare subsequent frames and only makes adjustments in 5-mV intervals (limit of electronics, corresponds to 1-2 pixels on the cameras)
- Exception handling blocks mean the scripts run continuously without user input



Conceptual layout of the slow stabilization system communication

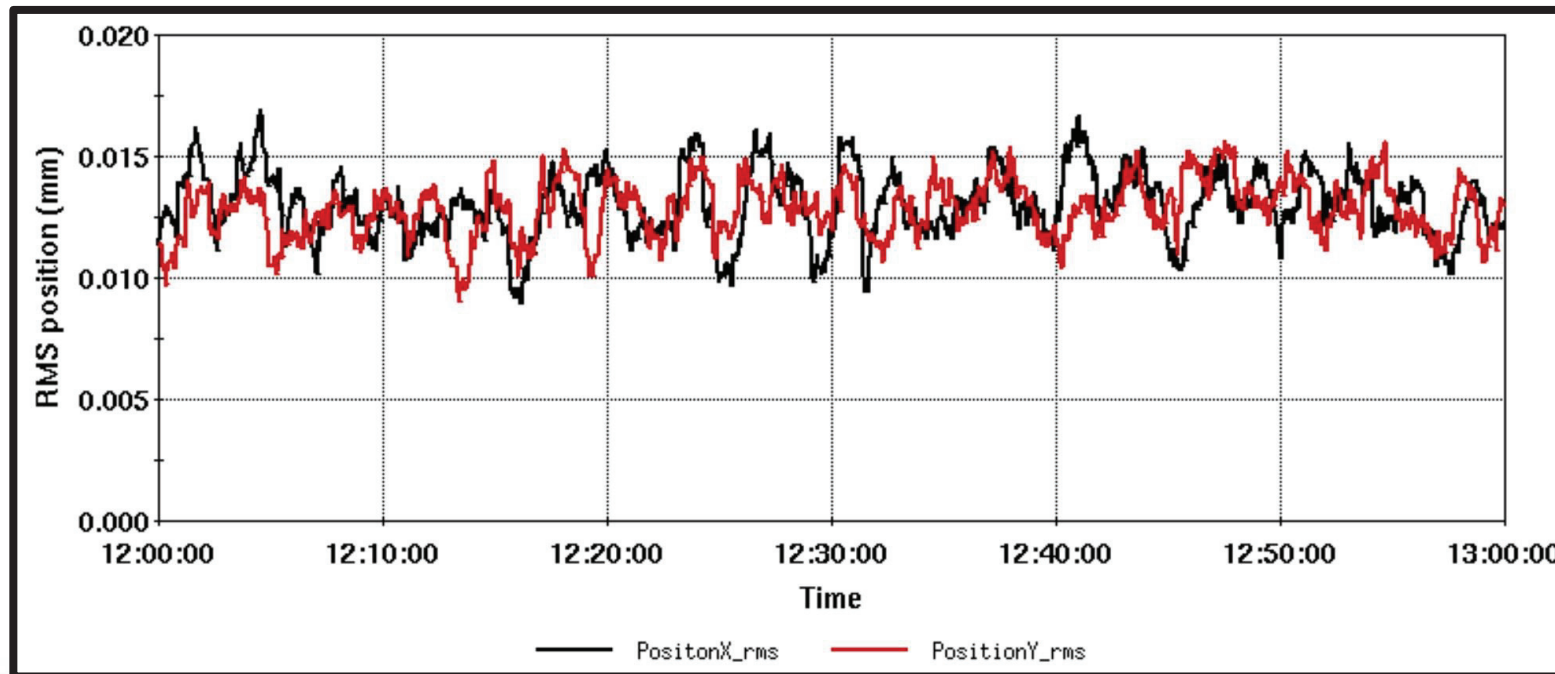


Simplified flow diagram of slow stabilization system for the gun table
(more details in backup slides)

Results from Operations (2019 Run)

Fast Stabilization System

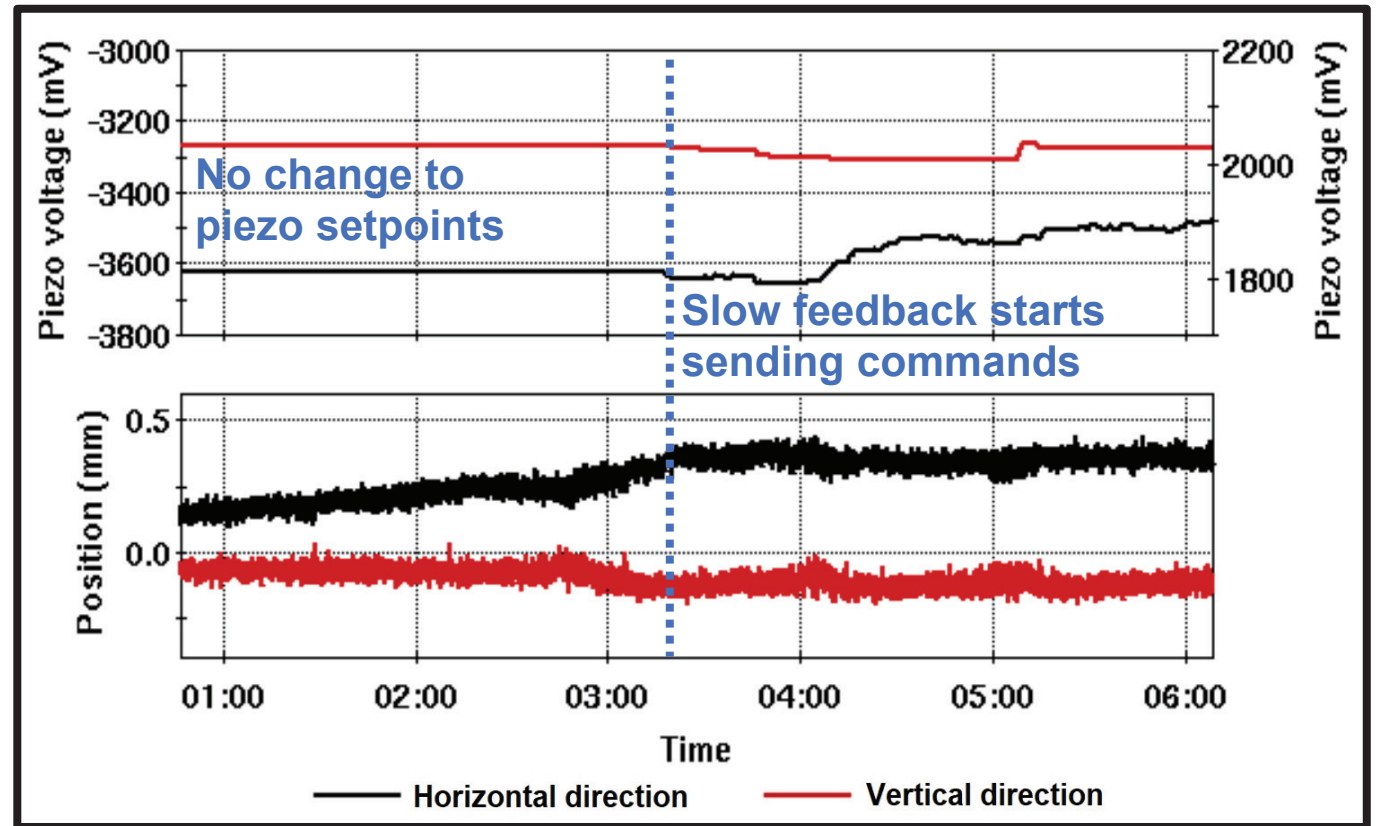
When available in CW mode, the activation of the fast stabilization system succeeded in providing rms position stability near 10 μm rms with optimized alignment. However, values between 15 and 20 μm rms were more typical.



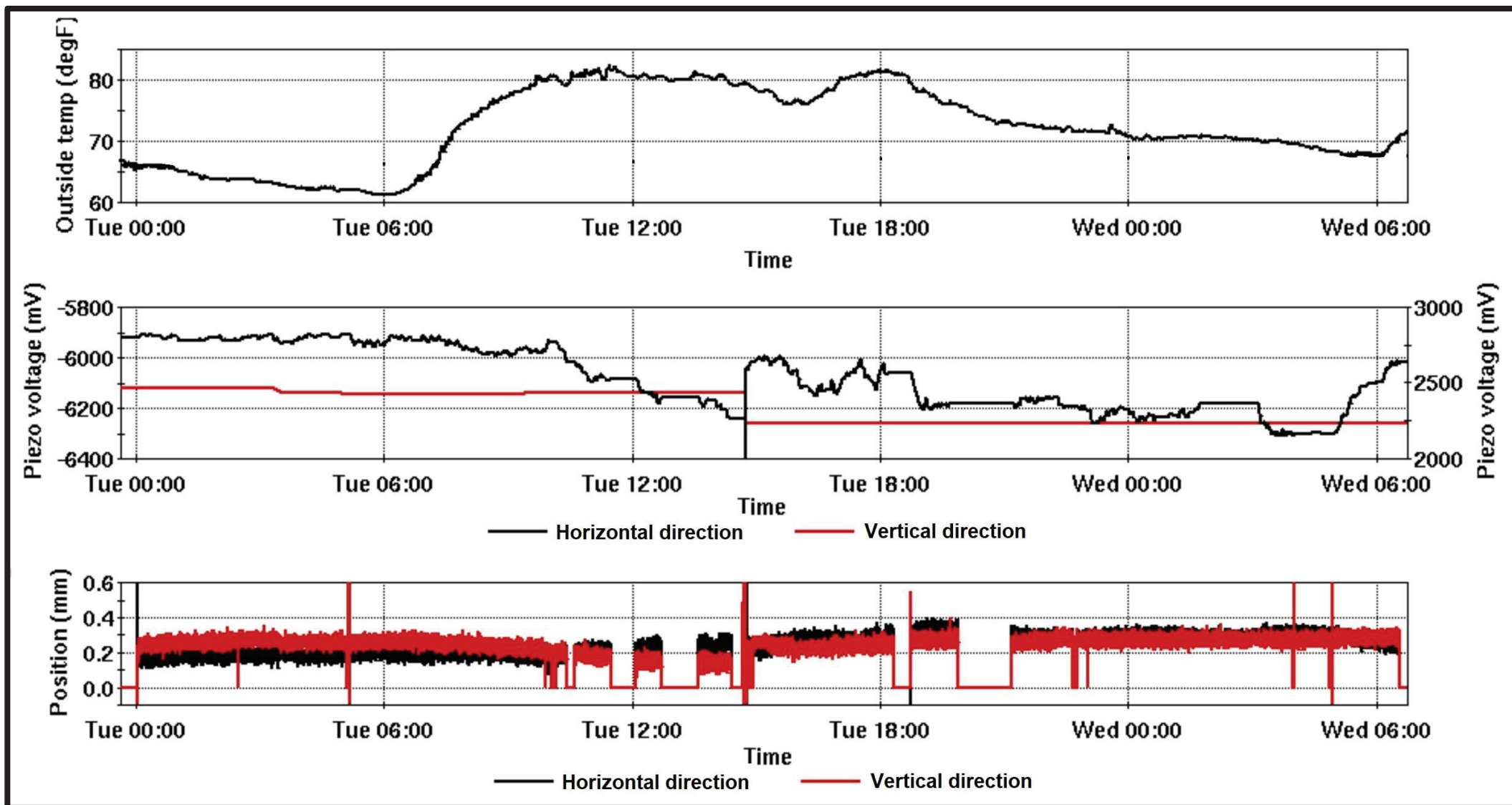
One-hour plot showing rms position variation near 10 μm rms
with fast stabilization system active

Slow Stabilization System

Effect on beam position (bottom plot) once commands are allowed to be sent to the piezo mirrors by the system (top plot)



- As anticipated, the slow active stabilization system was available in all laser modes and required no user input during the run
- System also achieved theoretical limit of ~10-micron rms position variation (limit was determined during proof-of-concept testing with an alignment laser and 70-meter transport)
 - However, standard performance from the slow stabilization system is currently considered 20 μm rms due to the increased sensitivity to structural changes in the laser beam profile



Position (bottom), piezo commands (middle), and outside temperature (top) over a 30-hour period in pulsed mode, showing achievement of theoretical limits of slow stabilization ($\sim 10\text{ }\mu\text{m rms}$)

Conclusion

After being implemented during the 2019 Run, these active pointing stabilization techniques largely yielded the expected results and contributed to the LEReC project's success in achieving bunched beam cooling.

Future efforts will focus on achieving consistency in both fast and slow stabilization systems as LEReC moves forward into sustained operations for RHIC Physics.

Thank you